

Samyak Rawlekar

657-273-2442 samyagr2@illinois.edu Homepage Google Scholar LinkedIn

Education

University of Illinois Urbana-Champaign

Ph.D. in Electrical and Computer Engineering

Advisor: Prof. Narendra Ahuja

Illinois, USA

Aug. 2023 – May. 2028

New York University

MS in Computer Engineering

Advisors: Prof. Yao Wang, Prof. Sumit Chopra

New York, USA

Sept. 2021 – May 2023

Indian Institute of Technology, IIT Dharwad

Bachelor of Technology in Electrical Engineering

Dharwad, India

Aug. 2017 – May 2021

Professional Experience

Amazon | Applied Scientist Intern

May 2026 – Aug 2026

Multimodal Composed Document Retrieval | Advisors: Dr. Yuanwei Wu, Dr. Hongcheng Wang

Seattle, WA

- Created CDocIR, the first composed document image retrieval benchmark (22,925 triplets) specifically designed to evaluate localized layout, content, and style modifications
- Developed EditPointer that replaces global embeddings with text-conditioned local patch prediction to explicitly isolate fine-grained visual changes among near-duplicate documents.
- Designed an edit-guided algorithm to prune redundant region comparisons, achieving a 3.8x inference speedup

Waymo | Perception Intern

May 2025 – Aug 2025

End-to-End Foundational Models | Advisors: Dr. Han Deng, Dr. Charlie Xu

Mountain View, CA

- Designed a **Latent Flow Matching (LFM)** framework for efficient trajectory prediction, achieving accuracy comparable to multi-modal large language models with reduced compute.
- Investigated conditioning mechanisms using **Gemma-3 MLLMs** to guide flow generation and improve prediction accuracy.

New York University | Research Intern

June 2022 – Sept 2022

Multi-Modal Representation Learning | Advisors: Dr. Cem Deniz, Dr. Sumit Chopra

Manhattan, NY

- Developed a joint weakly and self-supervised representation learning model for medical images.
- Proposed a novel fusion approach of images and clinical reports from the MIMIC-CXR dataset.

Publications

Peer-reviewed Conference Papers

- Complementing DINO Features with Image Structure for Part Discovery *NeurIPS 2026*
Samyak Rawlekar, Nikhil Chowdary Paleti, Amey Gupta, Narendra Ahuja
- Finding Distributed Object-Centric Properties in Self-Supervised Transformers *CVPR 2026 - Highlight (top-3%)*
Samyak Rawlekar, Amitabh Swain, Yujun Cai, Yiwei Wang, Ming-Hsuan Yang, Narendra Ahuja
- Efficiently Disentangling CLIP for Multi-Object Perception *Preprint*
Samyak Rawlekar, Yujun Cai, Yiwei Wang, Ming-Hsuan Yang, Narendra Ahuja
- Rethinking Prompting Strategies for Multi-Label Recognition with Partial Annotations *WACV 2025*
Samyak Rawlekar, Shubhang Bhatnagar, Narendra Ahuja
- Improving Multi-label Recognition using Class Co-Occurrence Probabilities *CVPRW 2024 & ICPR 2024 - Oral (top-5%)*
Samyak Rawlekar*, Shubhang Bhatnagar*, VP Srinivasulu, Narendra Ahuja
- A Dual-Phase Approach for Reconstructing Dynamic Shape and Skeleton of Articulated Objects from Monocular Video *ICML 2024*
Hao Zhang, Fang Li, **Samyak Rawlekar**, Narendra Ahuja
- Learning Implicit Representation for Reconstructing Articulated Objects *ICLR 2024*
Hao Zhang, Fang Li, **Samyak Rawlekar**, Narendra Ahuja
- Feature Compression for Rate Constrained Object Detection on the Edge *IEEE MIPR 2022*
Zhongzheng Yuan, **Samyak Rawlekar**, Siddharth Garg, Elza Erkip, Yao Wang
- Radiology Reports Improve Visual Representations Learned from Radiographs *MIDL 2023*
Haoxu Yuan, **Samyak Rawlekar**, Sumit Chopra, Cem Deniz

Journal Articles

- Split Computing With Scalable Feature Compression for Visual Analytics on the Edge *IEEE Trans. on Multimedia 2024*
Zhongzheng Yuan, **Samyak Rawlekar**, Siddharth Garg, Elza Erkip, Yao Wang
- Improving Performance of DL Predictive Models for COVID by Incorporating Environmental Parameters *Gondwana Research 2022*
Roshan Wathore, **Samyak Rawlekar**, Saima Anjum, Ankit Gupta, Hemant Bherwani, Nitin Labhasetwar, Rakesh Kumar

Teaching and Research Positions

- **Research Assistant** — CVRL, UIUC (with Prof. Narendra Ahuja) Aug 2023 – Present
- **Research Mentor** — PURE Program, UIUC Aug 2023 – Dec 2023
- **Research Assistant** — CILVR Lab, NYU (with Prof. Sumit Chopra) Sept 2022 – May 2023
- **Research Assistant** — Video Lab, NYU (with Prof. Yao Wang) Sept 2021 – May 2023
- **Teaching Assistant** — Image and Video Processing, NYU Jan 2023 – May 2023
- **Teaching Assistant** — Machine Learning, IIT Dharwad Aug 2021 – Nov 2021

Awards & Fellowships

- **Joan and Lalit Bahl Fellowship** — ECE Department, UIUC Aug 2024 – May 2026
- **Academic Fellowship** — ECE Department, UIUC Aug 2023 – Jul 2024
- **Academic Fellowship** — Fully-Funded Masters Program, NYU Sept 2021 – May 2023

Technical Skills

- **Languages:** Python, MATLAB, C/C++
- **Frameworks & Tools:** PyTorch, JAX, Linux, Vim, Slurm

Professional Service

- **Reviewer:** ICLR, ICML, WACV, NeurIPS, CVPR, AAAI